Energy-Efficient Service Migration for Multi-User Heterogeneous Dense Cellular Networks

Xiaobo Zhou, *Senior Member, IEEE*, Shuxin Ge, Tie Qiu, *Senior Member, IEEE*, Keqiu Li, *Senior Member, IEEE*, and Mohammed Atiquzzaman, *Senior Member, IEEE*

Abstract—Mobile edge computing (MEC) is a key enabler for ultra-low latency in heterogeneous dense cellular networks in the 5G era and beyond, by deploying services at the network edge. Due to high user mobility, the services are usually migrated to follow the users by predicting the user trajectory to achieve a balance between energy consumption and service latency. However, service migration for multi-user heterogeneous dense cellular networks is challenging because (1) the user trajectory prediction, which is crucial for service migration, becomes intractable with a large number of users, and (2) making service migration decisions for each user independently is subjected to interference among the users. Therefore, in this study, we formulated the service migration of all the users in MEC-enabled heterogeneous dense cellular networks as an optimization problem, with the objective of minimizing the average energy consumption while satisfying the service latency requirements, taking into account the interference among different users. Next, we developed an efficient energy-efficient online algorithm based on the Lyapunov and particle swarm optimizations, called EGO, to resolve the original problem without predicting the trajectories of the users. Finally, a series of simulations based on real-world mobility traces of vehicles in Bologna were conducted to establish the superiority of the EGO algorithm over state-of-the-art solutions.

Index Terms—Mobile edge computing, service migration, multi-user, Lyapunov optimization, particle swarm optimization

1 Introduction

Mobile Edge Computing (MEC) is emerging as a promising paradigm for future mobile networks, which pushes computing and storage resource to the edge of the network to improve the quality of networks performance [1], [2], [3], [4]. The development of MEC promotes an explosive growth in innovative delay-sensitive services, such as augmented reality, video streaming, and online gaming [5]. To satisfy the stringent latency requirement of these services, 5G networks forms a heterogeneous dense cellular networks, where the density reaches 40/50 base stations (BSs) equipped with edge servers per square kilometer [6]. The services are deployed at the BSs based on an existing virtual machine (VM) technique such that the requests from users can be served locally. However, due to the limitation of resource of edge servers, only a portion of services can be deployed at each edge server [7]. For a BS, if a user requests a service that has not been deployed locally, a possible solution is to route this request to nearby BSs or to the remote cloud, whereby the response delay is increased.

Service caching is utilized to ensure all the users can enjoy the ultra-low latency performance of MEC by determining the set of services deployed at each edge server [8], [9]. Generally, the service application maintains an internal state for each user, such as intermediate data processing results or history of data. Therefore, taking the request statistics into account, we can make a suitable caching strategy to effectively reduce the latency and energy consumption caused by routing the request [10]. In multi-user heterogeneous dense cellular networks, service caching may become invalid due to the dynamic requests introduced by user mobility. As the user moves among the BSs, the connected BS of user changes with received signal strength indication, called signal handoff. After signal handoff, the energy consumption of request routing as well as response time may increase and even becomes unacceptable. Although we can frequently update the service caching policy, the resource limitation in backhaul links results in additional latency and energy consumption during edge-cloud transmission process [11].

Service migration[1] triggered by user mobility is efficient in reducing the edge-cloud latency and energy consumption [12]. Migration helps services to follow the user,[2] which reduces request routing latency [13], [14]. However, more energy will be consumed in the service migration process. Existing researches focus on designing a migration strategy from the perspective of a single user, which always selects

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1. Note that signal handoff and service migration are different concepts. After signal handoff, the services may not be migrated to the new BS.

2. Note that service migration will not occur in the new Cloud Radio Access Network (Cloud-RAN), as the user requests service from a centralized baseband pool. However, Cloud-RAN is still in its infancy and many challenges have to be solved before it is widely deployed. Therefore, in this paper, we focus on service migration in conventional cellular networks enabled by mobile edge computing.

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the optimal edge server node by predicting the user trajectory based on Markov decision process (MDP) model [15].

In multi-user heterogeneous dense cellular networks, a feasible option is to apply the single-user service migration strategy independently for each user, in which case two key challenges must be addressed. First, the uncertain interference among users, which is referred to as resource conflict between different users when they make service migration decisions, may lead the migration to fail. Since the service migration decision is made by each user independently, there may be multiple users migrate service to the same BS, where the total resource requirement is far beyond the resource limitation of the BS. Moreover, the migration strategy for a service is shared by multiple users, which is not taken into account in the single-user service migration strategies, leading to resource scarcity and abuse [16]. Second, accurately predicting the trajectory plays a critical role in ensuring the efficiency of the service migration strategy. The existing mobility management schemes do not take the duration time that the user connects to target BS into account. Meanwhile, since the state space explosion occurs with high number of users, it is difficult to accurately predict the trajectory of each user [17].

Meanwhile, energy consumption in multi-user heterogeneous dense cellular networks becomes more and more important. According to the reports from MTN Consulting company and Huawei [18], [19], the energy consumption of a 5G BS is twice or more times than that of a 4G BS. It is expected that the power consumption will take up 18 percent of operational expenses in Europe, and 32 percent in India [20]. In heterogeneous dense cellular networks, service migration may happen more frequently due to the high density of BSs, leading to huge energy consumption. At present, energy consumption has gained considerable attention in both industry and academia, where an energy-efficient migration strategy is urgently required.

In this paper, we propose a novel **E**nergy-efficient service mi**G**ration for multi-user hter**O**geneous dense cellular networks (EGO) algorithm, which aims to minimize the total energy consumption while guaranteeing the service latency requirements and resource constraints. Note that the interference among users has been taken into account. The main contributions of this work are summarized below:

- An efficient multi-user service migration scheme for heterogeneous dense cellular networks is developed, and the service migration process is formalized as a **M**ixed-**I**nteger **N**onlinear **P**rogramming (MINP) problem.
- We propose EGO algorithm, which utilizes the Lyapunov optimization technique to decouple the migration process as a series of online sub-problems. EGO adaptively adjusts the resilient resource bounds for each service to efficiently tackle the interference among users. In each time slot, EGO makes migration decisions based on the difference-value between the expected energy consumption of migration and actual energy consumption of migration without foreseeing the trajectory.
- We make two modifications to the traditional PSO algorithm to solve the sub-problems. First, we design an updating rule based on probability of migrating service for each BS to improve the exploration of migration decision, and thus avoid converging to the local optimum. Second, we accelerate the convergence speed by using a correction mechanism to reduce the search space. The convergence and optimality of EGO is theoretically analyzed.
- The simulation results, which is established on the basis of real-world vehicle trajectory in Bologna, demonstrates the superior performance of EGO, especially with a large user number.

The rest of this paper is organized as follows: we investigate and summarize the related work in Section 2. The system overview, model and problem formulation considered in this study with formulation is proposed in Section 3. Next, the proposed EGO algorithm based on Lyapunov optimization and PSO is present in Section 4 with a detailed theoretical analysis. Section 5 describes the simulation environment, as well as parameters, and analyzes the experimental results. Finally, the conclusions are stated in Section 6.

2 Related Work

In MEC, services are deployed on the edge servers rather than the remote cloud to reduce the energy consumption and response latency [21], [22]. Owing to the resource limitation in each edge server and user mobility, a crucial issue of MEC that has attracted a considerable amount of attention is the efficient management of the services deployed at the edge servers (i.e., when and where to cache or migrate services) to meet the stringent latency requirements [23]. Various service management schemes were proposed, which can be classified into two categories namely, service caching and service migration.

Service caching (service placement) is similar to the VM caching in cloud computing, aiming to maximize the service hit rate by caching (placing) the services in the resource-limited edge servers while considering the request statistics of the users [11]. Chen *et al.* [24] proposed a collaborative service placement scheme to optimize the utility of MEC, which allows BSs to choose "to cooperate" or "not to cooperate" based on the graph coloring theory. Poularakis *et al.* [25] adopted a random rounding technique to develop a two-criterion algorithm for joint service placement to maximize the number of requests served by the edge servers or equivalently, minimize the load of the cloud. Xu *et al.* [10] presented a dynamic service caching method based on the Lyapunov optimization, where the objective was to reduce the average service latency, given the constraints on energy budget and storage space. Pu *et al.* [26] used regularization, rounding, and decomposition techniques to reduce the system costs over time to deal with the time-varying user requests. A drawback of these service caching methods is that the user mobility is not taken into account, leading to a reduction in the service hit rate as users move among the coverage area of different edge servers. Although this can be alleviated by frequently updating the service caching policy, significant edge-cloud transmission overheads would be incurred causing deterioration in the latency.

To account for user mobility, service migration balances energy consumption and latency adaptively by migrating the service to nearby BSs, in which case the mobility management

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and trade-off between energy consumption and latency usually are taken into account [8], [27]. For the mobility management in service migration, many solutions based on software definition networks (SDN) had been developed to improve the mobility management performance in 5G dense networks according to the radio signal strength [28]. Oliva *et al.* [29] presented a resilient SDN framework by building a virtual registration area to avoid updating users' location. In this framework, the local mobility anchor establishes a bidirectional tunnel between mobility access gateways located in edge servers, and thus achieve seamless handoff. Tartarini *et al.* [30] proposed a quality of service handoff rerouting component to generate route path before the users arrive at the new location using mobility prediction techniques. However, since the data required to be transmitted (e.g., application and history data) is much larger than session-based and resilient signal messages, addition latency will be caused. Moreover, the services may not be migrated to the other BS after the connected BS of user changes, which further complexes the service migration problem.

Thus, service migration should further take the migration energy consumption and latency performance into account. For a one-dimensional (1-D) mobility scenario, i.e., users can only go forward or backward, Ksentini *et al.* [31] proposed a method to predict user trajectory according to the MDP model, and then determine the process of service migration. Taking the more realistic situation. i.e., a two-dimensional (2-D) mobility scenario, into consideration, Wang *et al.* [15] made the migration decisions by predicting the trajectory based on a 2-D MDP model whose the state space is far larger than 1-D MDP model. Plachy *et al.* [32] presented a migration method based on predicting the trajectory utilizing channel information to determine the most suitable migration path and keep the energy consumption in low level. Sun *et al.* [14] made migration decisions by a user-centric scheme based on multi-armed bandit theories to satisfy the long-term energy budget. Note that these existing researches designed the migration strategies from the perspective of a single user [33]. The service migration strategies, made by different users independently, did not consider the interference and service sharing between users, which led to a reduction in the latency with limited resource. Moreover, it was difficult to sustain the prediction accuracy of trajectory with great growth in user number.

3 System Model and Formulation

3.1 System Overview

As shown in Fig. 1, in multi-user heterogeneous dense cellular networks, there are many BSs equipped with edge servers. Since BSs are densely deployed, there is an overlap of their coverage regions. The mobile users are randomly distributed, and move around the coverage regions of multiple BSs. In general, user connects to the BS with the largest received signal strength indicator [34]. The BSs are connected to each other via wireless link, and thus users can access all the services through routing between BSs. As the user moves, the services may not be migrated to the other BS even if the connected BS of the user changes. Normally, migration consumes more energy than request routing, yet, reduces request routing latency.

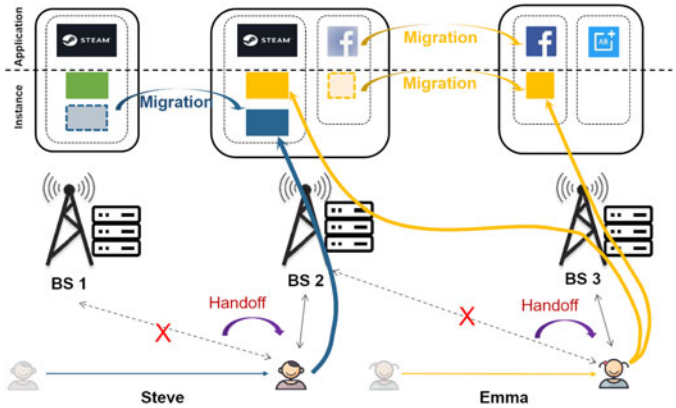

Fig. 1. An example of service migration in multi-user heterogeneous dense cellular networks.

The services are hosted by accessible tools that already exist in VM technique in edge server within limited resource. In this paper, the service is an abstraction of applications, which are deployed at the edge server using a 3-layer VM technique [35]. In previous work about service migration, the service is usually abstracted as a single Virtual Machine (VM). In this paper, we use a more realistic and fine-grained 3-layer VM model to abstract the service. Base layer is generic on each edge server as an operating system, application layer is shared by multiple users, and the instance layer of each user contains service state, e.g., request history, user information, and other private data. With the 3-layer VM technique, we assume the base layer has been deployed in each BS. A service request of the user must be responded by the BS which owns the application layer of this service and instance layer of this user. If the application layer has been deployed in target edge server, the user only needs to migrate its instance layer from previous serving node (an adjacent BS). Otherwise, the user needs to migrate both the application layer and the instance layer. Moreover, the 3-layer VM technique also applies to stateless services by setting the size of instance layer at 0. In other words, stateless service can be regarded as a special case of stateful service.

The operation of service migration is illustrated by a simple example shown in Fig. 1. In the system, there are three BSs and two users named Steve and Emma. At first, the services deployed at BS_1, BS_2, and BS_3 are $\{Steam\}$, $\{Steam\ and\ Facebook\}$, and $\{AR\}$, respectively. Emma and Steve are in the coverage area of BS_1 and BS_2, respectively. Emma generates requests to $Steam$, and Steve generates requests to $Steam$ and $Facebook$. As Steve and Emma move, Emma and Steve connect to BS_2 and BS_3, respectively. As $Steam$ is already deployed at BS_2, Emma migrates the corresponding instance layer to BS_2, and thus reduces the response latency. Meanwhile, Steve still requests $Steam$ from BS_2 by routing the request. Alternatively, both the application and instance layers of $Facebook$ are migrated to BS_3, such that Steve can be served locally.

In multi-user heterogeneous dense cellular networks, to minimize the average energy consumption while guaranteeing the latency requirements of the services, we have to answer the following questions:

- Which service should be migrated to which BS while considering the mobility of all the users, the resource

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TABLE 1
Definitions of Notations

Notation	Definition
$x_t^c(u,n)$	The connected BS of u in time slot t
$x_{m,t}(u,n)$	The serving node of the request of users u to service m in time slot t
$x'_{m,t}(u,n)$	The migration decision of user u to service m in time slot t
$\boldsymbol{P}_t$	The service application layer placement in time slot t
$\boldsymbol{P}'_t$	The service application layer placement in time slot t after migration
S_n	the storage capacity in BS n
λ_m	The input data size of service m
γ_m	The computation intensity of service m
D_m	The response deadline of service m
$f_{m,u}$	The CPU cycle requirements of a service m request from user u
θ_m^A	The application layer storage requirements of a service m
$\theta_{m,u}^I$	The instance layer storage requirements of a service m requested from user u
$W_{m,u}$	The channel bandwidth requirements of a service m requested from user u
$R(a,b)$	The transmit rate between source node a and target node b
$L_{m,t}(u)$	The service latency of user u for service m in time slot t
$E_{m,t}$	The total energy consumption for service m in time slot t

heterogeneity of all the BSs, and the interference between users?

- How to migrate the service while considering the 3-layer VM technique, i.e., to migrate the application layer together with the instance layer, or only to migrate the instance layer?

3.2 Service Placement Model

We consider a multi-user heterogeneous dense cellular networks consisting of N BSs with edge servers (indexed by n) and U mobile users (indexed by u). M services (indexed by m) can be deployed in the edge servers to serve users, such as video streaming, augmented reality and online gaming. The total time is divided into T slots, where each time slot lasts for τ seconds. In time slot t $(t = 0, 1, 2, \ldots, T)$, all the users generate service requests according to certain stochastic statistics. Table 1 summarizes the key parameter notations in our paper.

We use a U-by-N matrix $\boldsymbol{x}_t^c = \{x_t^c(u,n) | u = 1, 2, \ldots, U;\ n = 1, 2, \ldots, N\}$ to denote the connectivity between the user u and BS n in time slot t, where $x_t^c(u,n) = 1$ means the user u connects to BS n in time slot t, and $x_t^c(u,n) = 0$ indicates otherwise. Therefore, we have

$$\sum_{n=1}^{N} x_t^c(u,n) = 1, \forall t, u. \tag{1}$$

At the beginning of each time slot t, the network operator manages the service for each user u to decide which edge server n will provide service m to user u in the time slot $t+1$. We define a U-by-N binary matrix $\boldsymbol{x}_{m,t} = \{x_{m,t}(u,n) | u = 1, 2, \ldots, U;\ n = 1, 2, \ldots, N\}$, where $x_{m,t}(u,n) = 1$ indicates that the request of user u to service m is served by the server n in the time slot t, and $x_{m,t}(u,n) = 0$ indicates otherwise. Another U-by-N binary matrix $\boldsymbol{x}'_{m,t} = \{x'_{m,t}(u,n) | u = 1, 2, \ldots, U;\ n = 1, 2, \ldots, N\}$ is used to represent the migration decision in each time slot t, where $x'_{m,t}(u,n) = 1$ signifies that the service m of user u is migrated to edge server n in the time slot t, and $x'_{m,t}(u,n) = 0$ indicates otherwise. Thus, the server node is

$$x_{m,t+1}(u,n) = x'_{m,t}(u,n). \tag{2}$$

Since the request of user u to service m can be served by only one server, we have

$$\sum_{n=1}^{N} x'_{m,t}(u,n) \leq 1, \forall m, t, u. \tag{3}$$

As there is a one-to-one mapping between the user and its instance layer data, $x_{m,t}(u,n) = 1$ also indicates that the instance layer data of user u with service m is placed at edge server n in the time slot t.

However, due to the application layer is shared by multiple users, we use a M-by-N binary matrix $\boldsymbol{P}_t = \{P_t(m,n) | m = 1, 2, \ldots, M;\ n = 1, 2, \ldots, N\}$ to represent the edge service application layer placement state, where $P_t(m,n) = 1$ denotes that service m is placed in BS n in the time slot t, and $P_t(m,n) = 0$ indicates otherwise. Therefore, we have

$$P_t(m,n) = \min\left\{1, \sum_{u=1}^{U} x_{m,t}(u,n)\right\}, \forall m, t, n. \tag{4}$$

Similar to $\boldsymbol{P}_t$, $\boldsymbol{P}'_t$ is used to describe the service application layer placement state after migrating the service according to $x'_{m,t}(u,n)$ as follows:

$$P'_t(m,n) = \min\left\{1, \sum_{u=1}^{U} x'_{m,t}(u,n)\right\}, \forall m, t, n. \tag{5}$$

Since each diverse service has its demands, e.g., CPU cycles and maximum-execution time, let a 8-tuple $< \lambda_m, \gamma_m, D_m, f_{m,u}, \theta_m^A, \theta_{m,u}^I, W_m, \omega_m >$ to denote the service m.

- λ_m: the request data size,
- γ_m: the computation intensity (i.e., CPU Kcycles/bit required to compute for request),
- D_m: the response deadline,
- $f_{m,u}$: the requirement of CPU cycles of service for user u of service m,
- θ_m^A: the application layer data size for user u of service m,
- $\theta_{m,u}^I$: the instance layer data size for user u of service m,
- $W_{m,u}$: the channel bandwidth requirements for user u of service m

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- ω_m: the request frequency for service (i.e., requests to service m in each time slot).

To ensure that the decision of migration follows the computing, storage and channel bandwidth constraints for each BS, we have

$$\sum_{m=1}^{M}\sum_{u=1}^{U} f_{m,u} x'_{m,t}(u,n) \leq F_n, \forall t, n, \tag{6}$$

$$\sum_{m=1}^{M}\left[\sum_{u=1}^{U} \theta^I_{m,u} x'_{m,t}(u,n) + \theta^A_m P'_t(m,n)\right] \leq S_n, \forall t, n, \tag{7}$$

$$\sum_{m=1}^{M}\sum_{u=1}^{U} W_{m,u} x'_{m,t}(u,n) \leq W_n, \forall t, n, \tag{8}$$

where F_n, S_n and W_n are the maximum CPU frequency, the maximum storage size and the maximum channel bandwidth of the BS n, respectively, where n $(n = 1, 2, \ldots, N)$.

3.3 Service Latency

Let $R(a,b)$ denote the transmission rate between source node a and target node b. which is a non-linear function as follows:

$$R(a,b) = W\log_2\left(1 + \frac{p_a h d(a,b)^{-3}}{N_0}\right), \tag{9}$$

where p_a, N_0, W, h, $d(a,b)$ denote the transmission power of source node a, the noise power, the channel bandwidth, the complex fading vector, and the reference distance between source node a and the target node b, respectively [36].

When the user u generates a request to service m in time slot t, the transmission latency contains two parts. The first part is the latency that user sends the request to its connected BS. As the users are always on the move, it is difficult to obtain the transmission time between the user and its connected BS precisely. Therefore, we use a constant value C to approximate it. The second part is the latency caused by routing this request from user's connected BS to the user's serving BS. Thus, the transmission latency can be calculated by

$$l^{tra}_{m,t}(u) = \frac{\lambda_m}{R(\pi^c_{t,u}, \pi_{m,t,u})} + C, \tag{10}$$

where

$$\pi^c_{t,u} = \arg\max_n x^c_t(u,n), \tag{11}$$

$$\pi_{m,t,u} = \arg\max_n x_{m,t}(u,n), \tag{12}$$

and the computation latency is

$$l^{com}_{m,t}(u) = \frac{\lambda_m \gamma_m}{f_{m,u}}. \tag{13}$$

We use $\pi'_{m,t,u} = \arg\max_n x'_{m,t}(u,n)$ to denote the migration target BS, where the service m is migrated from BS $\pi_{m,t,u}$ to BS $\pi'_{m,t,u}$ in the time slot t. If the application layer of service m is already cached at BS $\pi'_{m,t,u}$, only the instance layer of user u needs to be migrated. Owing to the 3-layer framework of service, if the application layer of service m is not cached at BS $\pi'_{m,t,u}$, it can be cached at BS $\pi'_{m,t,u}$ from the nearest BS in advance without interrupting the ongoing service, and the migration latency is expressed as

$$l^{app}_{m,t}(n) = \min\left\{\frac{\theta^A_m |P'_t(m,n) - P_t(m,n)|}{P'_t(m,n') R(n,n')} | n' = 1, \ldots, N\right\}. \tag{14}$$

Hence, the service migration latency in the time slot t is the instance layer transmission latency, which is expressed as

$$l^{ins}_{m,t}(u) = \frac{\theta^I_{m,u}}{R(\pi_{m,t,u}, \pi'_{m,t,u})}. \tag{15}$$

$\pi_{m,t,u} = \pi'_{m,t,u}$ represents that the service m is not migrated in time slot t, i.e., the migration latency $l^{ins}_{m,t}(u) = 0$. Note that there may be more than one request in each time slot, let $L_{m,t}(u)$ denote the total service latency in time slot t, which is

$$L_{m,t}(u) = \omega_m\left[l^{tra}_{m,t}(u) + l^{com}_{m,t}(u)\right] + l^{ins}_{m,t}(u). \tag{16}$$

Since the request must receive response before deadline to ensure the latency, therefore, we have

$$L_{m,t}(u) \leq D_m, \forall u, m, t. \tag{17}$$

3.4 Energy Consumption

As a critical indicator, energy consumption comprises transmission energy

$$E^{tra}_{m,t} = p_{\pi_{m,t,u}} \omega_m \sum_{u=1}^{U} l^{tra}_m(u), \tag{18}$$

computing energy consumption

$$E^{com}_{m,t} = \kappa \omega_m \sum_{u=1}^{U} f^3_{m,u} l^{com}_{m,t}(u), \tag{19}$$

where κ is the unit energy consumption when edge server n is computing with one CPU cycle, and the total energy consumption of migration

$$E^{mig}_{m,t} = p_{\pi'_{m,t,u}}\left[\sum_{u=1}^{U} l^{ins}_{m,t}(u) + \sum_{n=1}^{N} l^{app}_{m,t}(n)\right]. \tag{20}$$

The data needed to be transmitted during the migration process is usually larger than that during the service request process. Thus, migration usually incurs more energy consumption and service latency. However, after migration, the energy consumption of transmitting service request, as well as service latency, will be greatly reduced in the following time slots. As a consequence, by properly making the migration decisions, the overhead induced by the migration can be compensated and performance gains in terms of energy consumption and service latency can be achieved.

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By summing these three parts, we can obtain the total energy consumption as

$$E_{m,t} = E_{m,t}^{tra} + E_{m,t}^{com} + E_{m,t}^{mig}. \tag{21}$$

3.5 Problem Formulation

The objective of this paper is to minimize the average energy consumption by selecting the optimal migration decisions $x'_{m,t}$, while the response deadline of each service and resource constraint in each BS can be satisfied. The problem can be formulated as follows:

$$\textbf{P1}: \quad \min_{\boldsymbol{x}'_{m,t}} \lim_{T\to\infty} \frac{1}{T}\sum_{m=1}^{M}\sum_{t=0}^{T-1} E_{m,t} \tag{22}$$

$$s.t. \quad l_{m,t}^{app}(n) \le \tau \qquad (1),(3),(6),(7),(8),(17). \tag{23}$$

Constraint (23) ensures that migration of the application layer will not interrupt the ongoing service. Let O^* denote the average energy consumption achieved by the optimal solution to **P1**. It is found that the lack of future information impedes the derivation of the optimal solution, i.e., optimally solving **P1** requires complete offline information such as user trajectory and request distribution, which is difficult to obtain. Moreover, even if the offline information is known, **P1** is an MINP problem that is proven to be NP-hard.

4 Online Service Migration Algorithm

In this section, we transform the long-term multi-service migration problem into a series of sub-problems by utilizing the Lyapunov optimization, and then solve the sub-problems with a modified PSO algorithm. We also show that under certain conditions, the optimal solutions of the sub-problems are equivalent to that of **P1**.

4.1 Problem Transformation

P1 can be divided into M sub-problems, i.e., making independent service migration for each service, which is

$$\textbf{P2}: \quad \min_{\boldsymbol{x}'_{m,t}} \lim_{T\to\infty} \frac{1}{T}\sum_{t=0}^{T-1} E_{m,t} \tag{24}$$

$$s.t. \quad (1),(3),(17),(23)\ \forall m$$

$$\sum_{u=1}^{U} f_{m,u}x'_{m,t}(u,n) = y_{n,t}^{f}(m) + \alpha_m(t,n)F_n, \tag{25}$$

$$\sum_{u=1}^{U} \theta_{m,u}^{I}x'_{m,t}(u,n) + \theta_m^A P'_t(m,n) = y_{n,t}^{s}(m) + \beta_m(t,n)S_n, \tag{26}$$

$$\sum_{u=1}^{U} W_{m,u}x'_{m,t}(u,n) = y_{n,t}^{w}(m) + \delta_m(t,n)W_n, \tag{27}$$

where $\alpha_m(t,n)$, $\beta_m(t,n)$ and $\delta_m(t,n)$ denote the fixed predicted proportion of the computation, storage and bandwidth resources of service m, which are

$$\alpha_m(t,n) = \frac{\omega_m \mathbf{E}\{f_{m,u}^2 | x'_{m,t-1}(u,n)=1,\forall u\}}{\sum_{m'=1}^{M}\omega_{m'}\mathbf{E}\{f_{m',u}^2 | x'_{m',t-1}(u,n)=1,\forall u\}}, \tag{28}$$

$$\beta_m(t,n) = \frac{\omega_m\left[\theta_m^A + \mathbf{E}\{\theta_{m,u}^I | x'_{m,t-1}(u,n)=1,\forall u\}\right]}{\sum_{m'=1}^{M}\omega_{m'}\left[\theta_{m'}^A + \mathbf{E}\{\theta_{m',u}^I | x'_{m',t-1}(u,n)=1,\forall u\}\right]}, \tag{29}$$

and

$$\delta_m(t,n) = \frac{\omega_m \mathbf{E}\{W_{m,u}^2 | x'_{m,t-1}(u,n)=1,\forall u\}}{\sum_{m'=1}^{M}\omega_{m'}\mathbf{E}\{W_{m',u}^2 | x'_{m',t-1}(u,n)=1,\forall u\}}, \tag{30}$$

respectively. Here, $y_{n,t}^{\star}(m), \star \in \{f,s,w\}$ are tuning factors. By setting proper values of the tuning factors, the resource occupied by service m can be accurately characterized such that the optimal solution of **P2** is equivalent to that of **P1**. We refer the right-hand side of (25), (26) and (27) as resilient resource bounds on computation, storage and bandwidth resources, respectively.

However, the exact value of $y_{n,t}^{\star}(m), \star \in \{f,s,w\}$ is difficult to be obtained ahead. Instead, we define virtual resource queues to record the total deviation from the predicted resource utilization based on Lyapunov optimization technique, which is

$$q_{n,t}^{\star}(m+1) = \max\{q_{n,t}^{\star}(m) + y_{n,t}^{\star}(m),\ 0\}, \star \in \{f,s,w\}. \tag{31}$$

The goal is to align the individual resource utilization of each service with that of all the services, such that less resource will be used for one service for the purpose of minimizing the energy consumption.

The Lyapunov function is defined as $L(x) \triangleq \frac{1}{2}x^2$, which indicates the "congestion level" in virtual resource queue. In this case, the quadratic Lyapunov function is

$$L(Q(m)) \triangleq \frac{1}{2T}\sum_{t=0}^{T-1}\sum_{n=1}^{N}\sum_{\star\in\{f,s,w\}}[q_{n,t}^{\star}(m)]^2. \tag{32}$$

The value of $L(Q(m))$ signifies the queue backlog, i.e., small value of $L(Q(m))$ implies that there is plenty of resource and virtual queue has a high stability. We introduce the one-slot conditional Lyapunov drift to shift the quadratic Lyapunov function towards a region of lower congestion, which is defined as

$$\begin{aligned}
\triangle_1^R(Q(m)) &\triangleq L(Q(m+1)) - L(Q(m)) \\
&\leqslant \frac{1}{2T}\sum_{t=0}^{T-1}\sum_{n=1}^{N}\sum_{\star\in\{f,s,w\}}[y_{n,t}^{\star}(m)]^2 \\
&\quad + \frac{1}{T}\sum_{t=0}^{T-1}\sum_{n=1}^{N}\sum_{\star\in\{f,s,w\}} q_{n,t}^{\star}(m)y_{n,t}^{\star}(m) \\
&\le B + \frac{1}{T}\sum_{t=0}^{T-1}\sum_{n=1}^{N}\sum_{\star\in\{f,s,w\}} q_{n,t}^{\star}(m)y_{n,t}^{\star}(m),
\end{aligned}$$

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where $B = \sum_{\star \in \{f,s,w\}} B_\star$

$$B_\star = \sup\left\{ \frac{1}{2T} \sum_{t=0}^{T-1} \sum_{n=1}^{N} [y_{n,t}^{\star}(m)]^2 \right\}, \star \in \{f, s, w\}. \tag{33}$$

By minimizing the quadratic Lyapunov function, the total resource utilization of each service in each BS can satisfy the resource bounds of each BS.

The objective of Lyapunov optimization is to minimize the one-slot drift-plus-penalty function, which is

$$\begin{aligned} \triangle_1^R(Q(m)) + \frac{V}{T} \sum_{t=0}^{T-1} E_{m,t} \leq \frac{V}{T} \sum_{t=0}^{T-1} E_{m,t} \\ + B + \frac{1}{T} \sum_{t=0}^{T-1} \sum_{n=1}^{N} \sum_{\star \in \{f,s,w\}} q_{n,t}^{\star}(m) y_{n,t}^{\star}(m). \end{aligned} \tag{34}$$

where $V > 0$ is to balance the energy consumption and the resources utilization. After constructing the resource virtual queue, **P2** is decomposed into many service-by-service sub-problems **P3** with resilient resource bounds. The objective is to devise a current resource allocation state to avoid a conflict between energy consumption and resource utilization for each service. The network operator determines migration decisions for each service m by solving **P3** as follows:

$$\begin{aligned} \textbf{P3}: \min_{\boldsymbol{x}'_{m,t}} \lim_{T \to \infty} \frac{1}{T} \sum_{t=0}^{T-1} \left[VE_{m,t} + \sum_{n=1}^{N} Q_{n,t}(m) \right] \\ s.t. \qquad (1), (3), (17), (23)\ \forall m, \end{aligned} \tag{35}$$

where

$$\begin{aligned} Q_{n,t}(m) = q_{n,t}^{f}(m) \sum_{u=1}^{U} f_{m,u} x'_{m,t}(u,n) \\ + q_{n,t}^{s}(m) \left[\sum_{u=1}^{U} \theta_{m,u}^{I} x'_{m,t}(u,n) + \theta_m^A P'_t(m,n) \right] \\ + q_{n,t}^{w}(m) \sum_{u=1}^{U} W_{m,u} x'_{m,t}(u,n). \end{aligned} \tag{36}$$

It is evident that the objective function of **P3** is a long-term energy consumption that involves future information, therefore **P3** can only be solved offline. However, the decision of service migration must be made instantaneously in each time slot without future information, i.e., an online solution is required. Therefore, we leverage the Lyapunov optimization technique to convert **P3** into per-time slot optimization sub-problems.

Let $\hat{e}_m(t)$ and $E_{m,t}^{mig}$ denote the expected and actual energy consumption of migration in time slot t, respectively. $\hat{e}_m(t)$ can be calculated as

$$\hat{e}_m(t) = \frac{1}{t} \sum_{t'=0}^{t-1} E_{m,t'}^{mig}. \tag{37}$$

Let $y_m^e(t) = E_{m,t}^{mig} - \hat{e}_m(t)$ denote the difference between the actual and expected energy consumption of migration in time slot t, and construct the energy queue (the initial queue backlog is 0) as follows:

$$e_m(t+1) = \max\{e_m(t) + y_m^e(t), 0\}. \tag{38}$$

Algorithm 1. EGO Algorithm

Input: $q_{n,t}^f(0) \leftarrow 0, q_{n,t}^s(0) \leftarrow 0, q_{n,t}^w(0) \leftarrow 0, e_m(0) \leftarrow 0,$
$\boldsymbol{x}_{m,0}, \boldsymbol{x}'_{m,0}, F_n, S_n, W_n, \boldsymbol{\alpha}_m, \boldsymbol{\beta}_m, \delta_m$
Output: service migration decision $\boldsymbol{x}'_{m,t}$

```
for t = 0 to T − 1 do
    update P_t according to (4);
    for m = 1 to M do
        update q^f_{n,t}(m), q^s_{n,t}(m) and q^w_{n,t}(m) according to (31);
        obtain x'_{m,t} by solving P4;
    end for
    update e_m(t + 1) according to (38);
    update x_{m,t+1} according to (2);
end for
```

We can see that $e_m(t) \gg 0$ indicates that more energy was consumed by migration in previous time slots, thus the energy consumption in time slot t, $E_{m,t}^{mig}$, should be smaller or even be set to 0 (i.e., do not migrate the service) in order to compensate then energy consumption such that the energy queue is stable. On the contrary, $e_m(t) \approx 0$ indicates we can use more energy for migration in time slot t.

The Lyapunov function for energy consumption of migration is

$$L(e_m(t)) = \frac{1}{2} [e_m(t)]^2. \tag{39}$$

Therefore, we can obtain the one-slot conditional Lyapunov drift to shift the quadratic Lyapunov function towards a region of lower congestion, which is defined and bounded as

$$\begin{aligned} \triangle_1^E(e_m(t)) \triangleq L(e_m(t+1)) - L(e_m(t)) \\ \leqslant \frac{1}{2} y_m^e(t)^2 + e_m(t) y_m^e(t) \leqslant B_e + e_m(t) y_m^e(t), \end{aligned} \tag{40}$$

where B_e is the upper bound of $\frac{1}{2} y_m^e(t)^2$. To stabilize the queue of energy consumption in migration while minimizing the objective function of **P3**, algorithm can be designed to make control actions that greedily minimize a supremum bound on the following drift-plus-penalty expression in each slot t

$$\begin{aligned} \triangle_1^E(e_m(t)) + V' \left[VE_{m,t} + \sum_{n=1}^{N} Q_{n,t}(m) \right] \\ \leqslant B_e + e_m(t) y_m^e(t) + V' \left[VE_{m,t} + \sum_{n=1}^{N} Q_{n,t}(m) \right] \\ = B_e + e_m(t) E_{m,t}^{mig} - e_m(t) \hat{e}_m(t) \\ + V' \left[VE_{m,t} + \sum_{n=1}^{N} Q_{n,t}(m) \right]. \end{aligned} \tag{41}$$

It can be observed B_e and $e_m(t)\hat{e}_m(t)$ are constant in each time slot. Therefore, under the Lyapunov frame work, the networks operator can determine migration decisions in each time slot by solving **P4** as follows:

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$$\begin{aligned} \mathbf{P4}: \min_{\boldsymbol{x}'_{m,t}} \quad & V'\left[VE_{m,t} + \sum_{n=1}^{N} Q_{n,t}(m)\right] + e_m(t)E^{mig}_{m,t} \\ s.t. \quad & (1), (3), (17), (23), \forall m, t, \end{aligned} \tag{42}$$

where the positive parameter V' is used to adjust the tradeoff between the objective function and the energy consumption of migration. Thus, the energy deficit queue maintained without foreseeing the future guides the base stations towards saving the energy consumption of migration, thereby enabling online decision making.

In summary, the details of the proposed EGO algorithm to solve **P1** are shown in Algorithm 1. At the beginning of time slot t, the networks operator must update the service deployment $\boldsymbol{P}_t$ (Lines 2). For each service, the resource queues should also update according to (31) (Lines 4). After this step, we can solve **P4** with a modified PSO algorithm, which is detailed in Section 4.2 (Lines 5). Next, we update the energy queue (Lines 6-7). Finally, generating the system state in next time slot according to the migration decisions of **P4** (Lines 8).

4.2 Modified PSO Algorithm

In this section, PSO algorithm, detailed in Algorithm 2, is modified to solve **P4**. For simplicity, we use $\bar{z}_{m,t}(\boldsymbol{x}'_{m,t})$ to represent the objective function value of **P4**.

Traditional PSO algorithm works by building a population (called a swarm) of candidate solutions (called particles). These particles $\overline{x}_k,\ k = 1, 2, \ldots, K$, move around in the search-space according to the following formulas. The velocity v_k of the particle k is guided by the best particle position p_k in the search-space as well as the the best group position g, which is

$$v_k^{i+1} = \omega v_k^i + c_1 r_1 (p_k^i - \overline{x}_k^i) + c_2 r_2 (g^i - \overline{x}_k^i), \tag{43}$$

where i is the index of iteration, ω is the inertia weight, c_1 and c_2 are the learning factors, and $r_1, r_2 \in [0, 1]$ are random numbers. It can be seen from (43) that $\boldsymbol{v}_k^{i+1}$ is a combination of three components, namely, the inertia component, which means that the particle k has a tendency to maintain its previous velocity; the cognition component, which signifies that the particle k has a tendency towards its local optimum position; and the society component, which means that the particle k has a tendency towards the global optimum position. Particle $\overline{x}_k^i$ moves as

$$\overline{x}_k^{i+1} = \overline{x}_k^i + v_k^i. \tag{44}$$

This process is repeated to eventually discover the satisfactory solution by finding the best fitness of particles. We use the value of $\bar{z}_{m,t}(\overline{x}_k^i)$ to denote the fitness of each particle i.e., a smaller value of $\bar{z}_{m,t}(\overline{x}_k^i)$ indicates a better fitness. Thus, we have (Lines 9-10)

$$p_k^i = \arg\min_{p_k^{i'}} \bar{z}_{m,t}(p_k^{i'}), i' = 1, 2, \ldots, i-1, \tag{45}$$

and

$$g^i = \begin{cases} g^{i-1}, & \bar{z}_{m,t}(\hat{p}_k^i) > \bar{z}_{m,t}(g^{i-1}), \\ \hat{p}_k^i, & \text{otherwise}. \end{cases} \tag{46}$$

where

$$\hat{p}_k^i = \arg\min_{p_k^i} \bar{z}_{m,t}(p_k^i), k = 1, 2, \ldots, K. \tag{47}$$

To find the best global position, each particle in each population must move in the solution space by tracking the best particle position $\boldsymbol{p}_k^i$ and best group position $\boldsymbol{g}^i$ until the criterion for minimum error is attained, i.e., $\max\{\bar{z}_{m,t}(p_k^i)\} - \bar{z}_{m,t}(g^i) < \varepsilon$, or the iterations reach a certain number.

In this scenario, the particle $\overline{\boldsymbol{x}}_{k,m,t}^i$ represents a potential optimal solution for service m in time slot t. $\overline{\boldsymbol{x}}_{k,m,t}^i$ is a U-by-N binary matrix, so we have $\overline{\boldsymbol{x}}_{k,m,t}^i \in \{0,1\}^{U \times N}$. Therefore, we can not update $\overline{\boldsymbol{x}}_{k,m,t}^i$ according to Formulas (43) and (44). We make two modifications to traditional PSO algorithm:

(1) We design a new updating rule for each particle, which can avoid converging into the local optimum. The velocity of $\overline{\boldsymbol{x}}_{k,m,t}^i$ can be updated by (Line 4)

$$\boldsymbol{v}_{k,m,t}^{i+1} = \frac{\omega \boldsymbol{v}_{k,m,t}^i + c_1 r_1 \boldsymbol{p}_{k,m,t}^i + c_2 r_2 \boldsymbol{g}_{m,t}^i}{\omega + c_1 r_1 + c_2 r_2}, \tag{48}$$

where $\omega + c_1 r_1 + c_2 r_2$ is used for normalization to ensure that

$$\sum_{n=1}^{N} v_{k,m,t}^i(u, n) = 1, \forall u. \tag{49}$$

In the tradition PSO algorithm, the particle always chooses BS $n^{*,i} = \arg_n \max v_{k,m,t}^i(u, n)$ as the target BS. The velocity $\boldsymbol{v}_{k,m,t}^{i+1}$ is updated according to (48). At the beginning of ith iteration, the best particle position $p_k^i(u, n)$ is equal to $v_{k,m,t}^i(u, n)$, and the influence of best group position $g^i(u, n)$ is weak. In this case, $n^{*,i+1}$ may always be equal to $n^{*,i}$, and thus the solution easily falls into local optimum.

To overcome this problem, we generate $\mathcal{F}_{k,m,t}^i(u)$ as a discrete probability distribution defined as

$$\mathcal{F}_{k,m,t}^i(u) \sim \begin{pmatrix} 1 & 2 & \cdots & N \\ v_{k,m,t}^i(u,1) & v_{k,m,t}^i(u,2) & \cdots & v_{k,m,t}^i(u,N) \end{pmatrix},$$

where $v_{k,m,t}^i(u, n)$ can be regarded as the probability of user u decides to migrate the service to BS n. We choose the target BS by generating $r_3 \sim \mathcal{F}_{k,m,t}^i(u)$ for each particle. Then, the particle $\overline{\boldsymbol{x}}_{k,m,t}^{i+1}$ is updated by (Line 5)

$$\overline{\boldsymbol{x}}_{k,m,t}^{i+1}(u, n) = \begin{cases} 1, & \text{if } n = r_3, \\ 0, & \text{otherwise}. \end{cases} \tag{50}$$

Note that it increases the exploration space to avoid falling into local optimum.

After updating the positions of the particles, we calculate the new fitness of all the particles, and update the individual optimal position $\boldsymbol{p}_k^i$ and global optimal position $\boldsymbol{g}^i$ by comparing with the previous optimal fitness according to Formulas (45) and (46).

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Algorithm 2. Modified PSO Algorithm for EGO

Input: $i \leftarrow 0, \boldsymbol{p}^0_{j,k}, \boldsymbol{g}^0_j, \overline{\boldsymbol{x}}^0_{j,k,m,t}, \boldsymbol{v}^0_{j,k}$
Output: service migration decision $\boldsymbol{x}'_{m,t}$
1: **while** $\max\{\bar{z}_{m,t}(\boldsymbol{g}^i_j)\} - \min\{\bar{z}_{m,t}(\boldsymbol{g}^i_j)\} < \epsilon$ **do**
2: **for** $j = 1$ to J **do**
3: **for** $k = 1$ to K **do**
4: update $\boldsymbol{v}^{i+1}_{j,k}$ according to (48);
5: update $\overline{\boldsymbol{x}}^{i+1}_{j,k,m,t}$ according to (50)
6: **if** $\mathbb{D}_m(t) > T_0$ **then**
7: update $\overline{\boldsymbol{x}}^{i+1}_{j,k,m,t}$ according to (52);
8: **end if**
9: update $\boldsymbol{p}^{i+1}_{j,k}$ according to (45)
10: update $\boldsymbol{g}^{i+1}_j$ according to (46)
11: **end for**
12: **end for**
13: $i++$;
14: **end while**
15: $\boldsymbol{x}'_{m,t} = g_j$

(2) We add a correction mechanism to improve the convergence speed. Let $\mathbb{D}(k,m,t)$ denote the euclidean distance between position of particle k for service m in time slot t and migration decisions for service m in time slot $t-1$ $\boldsymbol{x}'_{m,t-1}$, we have

$$\mathbb{D}(k,m,t) = \| \overline{\boldsymbol{x}}^i_{k,m,t} - \boldsymbol{x}'_{m,t-1} \| . \tag{51}$$

We notice that the migration decisions in time slot t is correlated with that in time slot $t-1$, therefore we can accelerate the convergence speed by decreasing the search space, i.e., keep the particles that satisfy $\mathbb{D}(k,m,t) \leq T_0$, where T_0 is a predefined threshold. For the particles with $\mathbb{D}(k,m,t) > T_0$, we set (Lines 6-8)

$$\overline{x}^{i+1}_{k,m,t}(u,n) = \begin{cases} 1, & n = \overline{n}(u), \\ 0, & \text{otherwise}, \end{cases} \tag{52}$$

where

$$\overline{n}(u) = \arg\max_n \overline{\boldsymbol{x}}^{i+1}_{k,m,t}(u,n) + \boldsymbol{x}'_{m,t-1}(u,n) + \boldsymbol{g}^i_{m,t}(u,n).$$

However, there is also little risk that the modified PSO algorithm may lead to a local optimal solution by always choosing a better decision in the combined optimizations. To avoid this situation, we extend the algorithm from a single population to multiple populations (J populations indexed by j). The iteration will stop when the criterion for minimum error between best group position in each population is attained, i.e., $\max\{\bar{z}_{m,t}(g^i_j)\} - \min\{\bar{z}_{m,t}(g^i_j)\} < \varepsilon$, or the iterations reach a certain number.

With increasing number of populations J, the decision has a greater probability to converge to the globally optimal solution with more iterations. Theorem 1 proves the convergence of the modified PSO algorithm.

Theorem 1. As $J > 0$ increases, the modified PSO algorithm converges with a higher probability to the globally optimal solution of **P4**. When $J \to \infty$, the algorithm converges to the globally optimal solution with probability 1.

Proof. The state of particle k in the ith iteration is denoted by $\boldsymbol{\varphi}^i_k = (\overline{\boldsymbol{x}}^{i+1}_{k,m,t}, \boldsymbol{v}^i_k, \boldsymbol{p}^i_k, \boldsymbol{g}^i)$. Evidently, $\{\boldsymbol{\varphi}^i_k, i \geq 1\}$ is a Markov chain, where the transition probability is [37]

$$\Pr(\boldsymbol{\varphi}^l_k|\boldsymbol{\varphi}^i_k) = \Pr(\overline{\boldsymbol{x}}^l_k|\overline{\boldsymbol{x}}^i_k).\Pr(\boldsymbol{v}^l_k|\boldsymbol{v}^i_k)\Pr(\boldsymbol{p}^l_k|\boldsymbol{p}^i_k)\Pr(\boldsymbol{g}^l|\boldsymbol{g}^i), \tag{53}$$

where

$$\Pr(\boldsymbol{p}^{i+1}_k|\boldsymbol{p}^i_k) = \begin{cases} 1, & \bar{z}_{m,t}(\boldsymbol{p}^{i+1}_k) \leq \bar{z}_{m,t}(\boldsymbol{p}^i_k)), \\ 0, & otherwise, \end{cases} \tag{54}$$

$$\Pr(\boldsymbol{g}^{i+1}|\boldsymbol{g}^i) = \begin{cases} 1, & \bar{z}_{m,t}(\boldsymbol{g}^{i+1}_k) \leq \bar{z}_{m,t}(\boldsymbol{g}^i_k)), \\ 0, & otherwise, \end{cases} \tag{55}$$

Similarly, let $\boldsymbol{\phi_k}^i = (\boldsymbol{\varphi_1}^i, \boldsymbol{\varphi}^i_2, \ldots, \boldsymbol{\varphi}^i_K)$ denote the state of population, $\{\boldsymbol{\phi}^i_k, i \geq 1\}$ is also a Markov chain. The transition probability is

$$\Pr(\boldsymbol{\phi}^l_k|\boldsymbol{\phi}^i_k) = \prod_{k=1}^{K} P(\boldsymbol{\varphi}^l_k|\boldsymbol{\varphi}^i_k). \tag{56}$$

We use $\boldsymbol{p}^*_k$ and $\boldsymbol{g}^*$ to represent the best particle and group positions, respectively. Thus, the optimal particle state is described by

$$\mathbb{K} = \left\{\boldsymbol{\varphi}^*_k = (\overline{\boldsymbol{x}}^i_k, \boldsymbol{v}^i_k, \boldsymbol{p}^*_k, \boldsymbol{g}^*), i \geq 1\right\}, \tag{57}$$

where $\mathbb{K}$ is a closed set. We also define the optimal population state set as

$$\mathbb{H} = \left\{\boldsymbol{\phi}^{i,*}, i \geq 1 | \boldsymbol{\phi}^{i,*} = (\boldsymbol{\varphi}^i_1, \boldsymbol{\varphi}^i_2, \ldots, \boldsymbol{\varphi}^i_K), \exists \boldsymbol{\varphi}^i_k \in \mathbb{K}\right\}, \tag{58}$$

which is also a closed set.

We define another closed set $\mathbb{B}$. It can be seen that $\mathbb{G} = \mathbb{H} \bigcup \mathbb{B}$ is a closed set. According to the law of total probability, the probability of $\phi^{i+1} \notin \mathbb{G}$ is

$$\begin{aligned} \Pr(\boldsymbol{\phi}^{i+1} \notin \mathbb{G}) = {} & \Pr(\boldsymbol{\phi}^{i+1} \notin \mathbb{G}|\boldsymbol{\phi}^i \notin \mathbb{G})\Pr(\boldsymbol{\phi}^i \notin \mathbb{G}) \\ & + \Pr(\boldsymbol{\phi}^{i+1} \notin \mathbb{G}|\boldsymbol{\phi}^i \in \mathbb{G})\Pr(\boldsymbol{\phi}^i \in \mathbb{G}). \end{aligned} \tag{59}$$

As $\mathbb{G}$ is a closed set, $\Pr(\boldsymbol{\phi}^{i+1} \notin \mathbb{G}|\boldsymbol{\phi}^i \in \mathbb{G}) = 0$, and we have

$$\begin{aligned} \Pr(\boldsymbol{\phi}^{i+1} \notin \mathbb{G}) &= \Pr(\boldsymbol{\phi}^{i+1} \notin \mathbb{G}|\boldsymbol{\phi}^i \notin \mathbb{G}) \cdot \\ &\quad \Pr(\boldsymbol{\phi}^i \notin \mathbb{G}|\boldsymbol{\phi}^{i-1} \notin \mathbb{G}) \ldots \Pr(\boldsymbol{\phi}^l \notin \mathbb{G}) \\ &= \Pr(\boldsymbol{\phi}^l \notin \mathbb{G}) \prod_{k=1}^{K} \Pr(\boldsymbol{\phi}^{l+1} \notin \mathbb{G}|\boldsymbol{\phi}^l \notin \mathbb{G}). \end{aligned} \tag{60}$$

When $\boldsymbol{\phi}^i = (\boldsymbol{\varphi}^i_1, \boldsymbol{\varphi}^i_2, \ldots, \boldsymbol{\varphi}^i_K) \notin \mathbb{G}$, $\forall k \in [1, K]$, $\boldsymbol{\varphi}^i_k \notin \mathbb{B}$ and $\boldsymbol{\varphi}^i_k \notin \mathbb{H}$. Thus, we have

$$\begin{aligned} \Pr(\boldsymbol{\phi}^{l+1} \notin \mathbb{G}|\boldsymbol{\phi}^l \notin \mathbb{G}) &= \prod_{k=1}^{K} \Pr(\boldsymbol{\varphi}^{l+1}_k \notin \mathbb{G}|\boldsymbol{\varphi}^l_k \notin \mathbb{G}) \\ &= \prod_{k=1}^{K} \Big[1 - \Pr(\overline{\boldsymbol{x}}^i_k|\overline{\boldsymbol{x}}^{i-1}_k)\Pr(\boldsymbol{v}^i_k|\boldsymbol{v}^{i-1}_k) \cdot \\ &\quad \Pr(\boldsymbol{p}^i_k|\boldsymbol{p}^{i-1}_k)\Pr(\boldsymbol{g}^i|\boldsymbol{g}^{i-1})\Big]. \end{aligned} \tag{61}$$

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By summing $\Pr(\boldsymbol{\phi}^i \notin \mathbb{G})$ over i, we have

$$\sum_{i=1}^{\infty} \Pr(\boldsymbol{\phi}^i \notin \mathbb{G}) = \Pr(\boldsymbol{\phi}^l \notin \mathbb{G}) \sum_{i=1}^{\infty} \prod_{l=1}^{i-1} \prod_{k=1}^{K} \Big\{1 - \Pr(\boldsymbol{g}^i | \boldsymbol{g}^{i-1}) \cdot \Pr(\overline{\boldsymbol{x}}_k^i | \overline{\boldsymbol{x}}_k^{i-1}) \Pr(\boldsymbol{v}_k^i | \boldsymbol{v}_k^{i-1}) \Pr(\boldsymbol{p}_k^i | \boldsymbol{p}_k^{i-1}) \Big\}. \tag{62}$$

As $\sum_{i=1}^{\infty} \Pr(\boldsymbol{\phi}^i) < \infty$, $\sum_{i=1}^{\infty} \Pr(\boldsymbol{\phi}^i \notin \mathbb{G}) < \infty$, we obtain

$$\lim_{i \to \infty} \Big\{1 - \Pr(\overline{\boldsymbol{x}}_k^i | \overline{\boldsymbol{x}}_k^{i-1}) \Pr(\boldsymbol{v}_k^i | \boldsymbol{v}_k^{i-1}) \cdot \Pr(\boldsymbol{p}_k^i | \boldsymbol{p}_k^{i-1}) \Pr(\boldsymbol{g}^i | \boldsymbol{g}^{i-1}) \Big\} = 0. \tag{63}$$

Therefore, we have

$$\lim_{i \to \infty} \Pr(\overline{\boldsymbol{x}}_k^i | \overline{\boldsymbol{x}}_k^{i-1}) \Pr(\boldsymbol{v}_k^i | \boldsymbol{v}_k^{i-1}) \cdot \Pr(\boldsymbol{p}_k^i | \boldsymbol{p}_k^{i-1}) \Pr(\boldsymbol{g}^i | \boldsymbol{g}^{i-1}) = 1.$$

This is equivalent to

$$\begin{aligned} &\lim_{i \to \infty} \Pr(\overline{\boldsymbol{x}}_k^i | \overline{\boldsymbol{x}}_k^{i-1}) = \lim_{i \to \infty} \Pr(\boldsymbol{v}_k^i | \boldsymbol{v}_k^{i-1}) \\ &= \lim_{i \to \infty} \Pr(\boldsymbol{p}_k^i | \boldsymbol{p}_k^{i-1}) = \lim_{i \to \infty} \Pr(\boldsymbol{g}^i | \boldsymbol{g}^{i-1}) = 1. \end{aligned} \tag{64}$$

By substituting (64) into (54) and (55), when $i \to \infty$, we have $\boldsymbol{p}_k^i = \boldsymbol{p}_k^{i+1}$ and $\boldsymbol{g}^i = \boldsymbol{g}^{i+1}$. Thus, we obtain $\Pr(\lim_{i \to \infty} \boldsymbol{\phi}^i \in \mathbb{G}) = 1$. This proves that $\{\boldsymbol{\phi}^i, i \geq 1\}$ converges to $\mathbb{G}$. When $\boldsymbol{g}^i = \boldsymbol{g}^{i+1} = \boldsymbol{g}^*$, $\{\boldsymbol{\phi}^i, i \geq 1\}$ converges to $\mathbb{H}$. This indicates that our algorithm converges to the globally optimal solution for a single population. However, when $\boldsymbol{g}^i = \boldsymbol{g}^{i+1} \neq \boldsymbol{g}^*$, $\{\boldsymbol{\phi}^i, i \geq 1\}$ cannot converge to $\mathbb{H}$. Let $\Pr(\boldsymbol{g}^i \neq \boldsymbol{g}^*)$ denote the probability of not converging to the global optimization for a population. The probability that at least one population converges to the global optimization when the population number is J is

$$\Pr(\boldsymbol{g}^* \in \mathbb{H}) = 1 - \big[\Pr(\boldsymbol{g}^i \neq \boldsymbol{g}^*)\big]^J. \tag{65}$$

When $J \to \infty$, $\lim_{J \to \infty} \Pr(\boldsymbol{g}^* \in \mathbb{H}) = 1$, which shows the probability to be 1 for global optimization, thereby completing the proof. □

4.3 Optimality Analysis

A process of proof and optimality analysis of EGO algorithm are shown in this section. We assume that there is at least one solution for **P1**. According to (31), we are easy to obtain

$$q_{n,t}^{\star}(m) - q_{n,t}^{\star}(m) \geq y_{n,t}^{\star}(m), \star \in \{f, s, w\}. \tag{66}$$

By summing (66) over m, we get

$$q_{n,t}^{\star}(M) - q_{n,t}^{\star}(0) \geq \sum_{m=1}^{M} y_{n,t}^{\star}(m), \star \in \{f, s, w\}. \tag{67}$$

According to (34), the drift for M services is

$$\triangle_M^R(Q(m)) \triangleq L(Q(M)) - L(Q(0)). \tag{68}$$

Therefore, we have

$$\begin{aligned} &\triangle_M^R(Q(m)) + \frac{V}{T} \sum_{m=1}^{M} \sum_{t=0}^{T-1} E_{m,t} \leq \frac{V}{T} \sum_{m=1}^{M} \sum_{t=0}^{T-1} E_{m,t} + MU \\ &+ \sum_{m=1}^{M} \frac{1}{T} \sum_{t=0}^{T-1} \sum_{n=1}^{N} \sum_{\star \in \{f,s,w\}} \Big[q_{n,t}^{\star}(m) - q_{n,t}^{\star}(0)\Big] y_{n,t}^{\star}(m), \end{aligned} \tag{69}$$

where $q_{n,t}^f(0) = 0$, $q_{n,t}^s(0) = 0$ and $q_{n,t}^w(0) = 0$. Let $y^{f*} \geq 0$, $y^{s*} \geq 0$ and $y^{w*} \geq 0$ denote the maximum of $y_{n,t}^f(m)$, $y_{n,t}^s(m)$ and $y_{n,t}^w(m)$, respectively. According to (67), we obtain

$$q_{n,t}^{\star}(m) - q_{n,t}^{\star}(0) \leq (m-1) y^{\star,*}. \tag{70}$$

Combined (70) with (69), we have

$$\sum_{m=1}^{M} \frac{1}{T} \sum_{t=0}^{T-1} \sum_{n=1}^{N} \sum_{\star \in \{f,s,w\}} \Big[q_{n,t}^{\star}(m) - q_{n,t}^{\star}(0)\Big] y_{n,t}^{\star}(m) \tag{71}$$

$$\begin{aligned} &\leq \sum_{m=1}^{M} \frac{1}{T} \sum_{t=0}^{T-1} N(m-1) \sum_{\star \in \{f,s,w\}} (y^{\star,*})^2 \\ &= \frac{NM(M-1)}{2T} \sum_{t=0}^{T-1} \sum_{\star \in \{f,s,w\}} (y^{\star,*})^2 = M(M-1)B. \end{aligned} \tag{72}$$

By substituting (71) into (69), we have

$$\triangle_M^R(Q(m)) + \frac{V}{T} \sum_{m=1}^{M} \sum_{t=0}^{T-1} E_{m,t} \leq VO^* + M^2 B. \tag{73}$$

According to [38], for any $\delta > 0$, there exist a stationary and randomized policy for **P4** that ascertains whether $\boldsymbol{x}'_{m,t}$ is independent of the current queue backlogs $e_m(t)$, such that the following inequalities are satisfied $\mathbf{E}[y_m^e(t)] \leq \delta$. Substituting (73) into (40), we have

$$\begin{aligned} &\triangle_1^E(e_m(t)) + V' \Big[\frac{\triangle_M^R(Q(m))}{M} + VE_{m,t}\Big] \\ &\leq B_e + V' \Big[VO^* + MB\Big] + \delta e_m(t). \end{aligned} \tag{74}$$

Let δ tend to 0, and summing (74) over t, we obtain

$$\begin{aligned} &\frac{\triangle_T^E(e_m(t))}{T} + \frac{V'}{T} \sum_{t=0}^{T-1} \left[\frac{\triangle_M^R(Q(m))}{M} + VE_{m,t}\right] \\ &\leq B_e + V'MB + V'VO^*. \end{aligned} \tag{75}$$

Next, by summing (75) over m, we obtain

$$\begin{aligned} &\sum_{m=1}^{M} \frac{\triangle_T^E(e_m(t))}{T} + \sum_{m=1}^{M} \frac{V'}{T} \sum_{t=0}^{T-1} \mathbf{E}\left[\frac{\triangle_M^R(Q(m))}{M} + VE_{m,t}\right] \\ &\leq \sum_{m=1}^{M} B_e + V'[M^2 B + VO^*]. \end{aligned} \tag{76}$$

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Dividing both sides of (76) by VV', the average energy consumption based on EGO algorithm satisfies

$$\sum_{m=1}^{M}\frac{1}{T}\sum_{t=0}^{T-1}E_{m,t}\leq\frac{1}{VV'}\sum_{m=1}^{M}B_e+\frac{M^2B}{V}+O^*. \tag{77}$$

According to (32) and (68), we obtain

$$2\triangle_M^R(Q(m))=\frac{1}{T}\sum_{t=0}^{T-1}\sum_{n=1}^{N}\sum_{\star\in\{f,s,w\}}[q_{n,t}^{\star}(M)]^2. \tag{78}$$

By substituting (78) into (67), and according to (73), we have

$$\frac{1}{T}\sum_{t=0}^{T-1}\sum_{n=1}^{N}\sum_{m=1}^{M}\sum_{\star\in\{f,s,w\}}[y_{n,t}^{\star}(M)]^2 \tag{79}$$

$$\begin{aligned}&\leq 2\triangle_M^R(Q(m))\leq 2(M^2B+VO^*)\\&=2M^2(B_f+B_s+B_w)+2VO^*\\&=\frac{1}{T}\sum_{t=0}^{T-1}\sum_{m=1}^{M}\sum_{n=1}^{N}\left[\frac{2M(B_f+B_s+B_w)}{N}+\frac{2VO^*}{NM}\right].\end{aligned} \tag{80}$$

Thus, we obtain

$$y_{n,t}^{\star}(M)\leq\sqrt{\frac{2MB_\star}{N}+\frac{2VO^*}{NM}},\star\in\{f,s,w\}. \tag{81}$$

By substituting (67) into (81), the computing, storage and channel bandwidth resource utilization satisfy

$$R_{n,t}^{\star,*}\leq F_n+\sqrt{\frac{2MB_\star}{N}+\frac{2VO^*}{NM}}^{\star}\in\{f,s,w\}. \tag{82}$$

As can be seen from (82), V' is not related to the resource constraints. By setting an appropriate V', the first term on the right-hand side of (77) becomes negligible. The average energy consumption of EGO algorithm, in the worst case, is not more than $O(1/V)$ with respect to O^*. Moreover, the resource is within a limited deviation $O(V)$ in comparison to the resource constraints. Therefore, through adjusting V, we can achieve a trade-off of $[O(1/V), O(V)]$ to balance the resource allocation and energy consumption. Note EGO is a centralized solution, however, an efficient distributed algorithm is preferred in large-scale networks. This is left as a future study.

5 Performance Evaluation

In this section, We use arcGIS to deal with the real-world vehicle trajectory collected in Bologna and randomly select 3000 vehicles as mobile users [39]. The sampling frequency of the trajectories is 1 Hz. We download the road map from OpenstreetMap, which is around 140 km^2. We proportionally scale down the original map to a 1500 m $\times$ 1060 m area served by 77 BSs according to the real communication tower location as shown in the Fig. 2. On the new map, the velocity of each user is around 1.4 m/s, which is quite close to the walking speed of human beings.

We use Matlab to run the simulations. In the simulations, users move according to the trajectory in each time slot, and request service from a BS where this service is deployed. As users move among the coverage area of different BS, the services can be migrated to another BS to serve users. If the service application layer has been deployed in target BS, only the instance layer needs to be migrated. Otherwise, the application layer can be migrated first while the service is running, then the service is suspended and the instance layer is transferred to the target BS. The channel is assumed to be the frequency-flat block-fading Rayleigh channel. The transmission power p, noise power N_0 are set to [0.1,0.4] W and 2×10^{-10} W, respectively. The value of complex fading vector h is modeled as circularly symmetric complex Gaussian random variables [40]. The performance metrics are the average energy consumption, average service latency, and deadline guarantee rate. Initially, we randomly select several services for each edge server. The parameters of services are shown in Table 2 [13], [41]. Each user requests some services that are independent of the other users. The computation resource and bandwidth of each service for each user, i.e., $f_{m,u}$ and $W_{m,u}$ follow uniform distribution [500, 1000] cycles/bit and [1,10] MHz, respectively. For each BS, the maximum resources F_n, S_n and W_n are generated randomly from [50,100] GHz, [1,5] GB and [0.5,1] GHz, respectively. We set $C=0.02$ s and $\kappa=10^{-26}$.

We compare the performance of the EGO algorithm with the three benchmarks given below:

- *Static optimization placement (SOP)*: This is a static service placement strategy, where the services are never migrated after they are placed at the edge servers.
- *Partial dynamic optimization algorithm (PDOA)* [33]: PDOA makes migration decisions with the objective of minimizing the average service latency from the perspective of a single user. In our simulations, each user uses PDOA independently.
- *Dynamic Markov decision process (DMDP)* [15]: This is also a single-user service migration algorithm, where the trajectory is predicted using the MDP model. The optimal service migration decision is made to minimize the energy consumption. Similarly, DMDP is also applied to each user independently in our simulations.

5.1 Impact of User Number

Fig. 3a shows the average energy consumption per user of the four methods, where the user number varies from 100 to 3000. It can be observed that the average energy per user of SOP is the highest among the four methods, this is because the services do no migrate as the user moves and thus more transmission energy is consumed. It is also found that the average energy per user with SOP monotonically increases as the user number increases, this is because more computing energy is consumed to ensure the service latency requirements of all the users. On the other hand, the average energy consumption per user of PDOA, DMDP and EGO first decreases as the user number increases, because the migration energy consumption of the application layer is shared by multiple users. However, after the user number exceeds a certain number, the average energy consumption per user increases, due to the fact that additional migration

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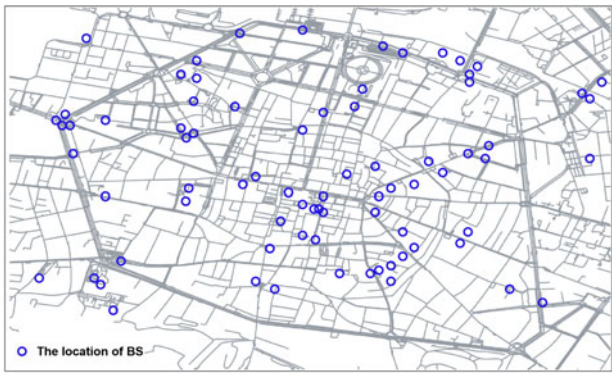

Fig. 2. The distribution of BSs.

TABLE 2
Service Parameters

Type	D_m (s)	λ_m (bits)	γ_m (cycles/bit)	ω_m	θ_m^A (MB)	θ_m^I (MB)
Emergency Stop	0.1	3200	36000	10	47	[1,3]
Collision Risk	0.1	4800	40000	10	70	[1,3]
Accident Report	0.5	4800	28000	2	97	[2,3]
Parking	0.1	1200	80000	1	85	[3,5]
Platoon	0.5	4800	88000	2	95	[4,6]
Face Detection	0.5	3200	50000	1	363	[5,10]
Video Stream	0.1	1200	10000	10	500	0

energy will be consumed due to user interference. It is interesting to find that when the number of users is smaller than 1000, the lowest average energy consumption is achieved with DMDP because of its objective is minimizing the average energy consumption. However, when the number of users is larger than 1000, the average energy consumption of EGO becomes the lowest as it adaptively adjusts the resilient resources bounds for each service to efficiently tackle the interference among users. Compared with DMDP, EGO saves 40.7 percent energy when there are 3000 users. It shows that EGO is suitable for multi-user heterogeneous dense cellular networks.

The average service latency per user of the four methods is depicted in Fig. 3b, where the user number varies from 100 to 3000. As the user number increases, the average service latency per user of all four methods increases. Since the services do no migrate and the user constantly moves, the additional service latency caused by request routing leads to the highest service latency of SOP. The average service latency of DMDP is the second highest as it focuses on minimizing the average energy consumption. PDOA achieves the lowest service latency when the user number is smaller than 1700. This is because the services are always migrated to follow the user. It is also found that the average service latency of PDOA grows exponentially when the user number exceeds 1200. This is because the interference among users becomes severe and dominates the performance. The performance of EGO is quite close to that of PDOA when user number is lower than 1800. When user number exceeds 1800, the lowest average service latency is achieved by EGO since it utilizes a resilient resource bounds to avoid the interference among users. Compared with PDOA, the service latency of EGO has a 12.6 percent gain with 3000 users.

Fig. 3c shows the average deadline guarantee rate, i.e., the percentage of requests that receive a response from the edge server before its deadline, of the four methods. Due to the long transmission path, SOP has the lowest average deadline guarantee rate. The average deadline guarantee rate of DMDP is the second lowest, and descends monotonously with the user number increases. Since the objective of PDOA is to minimize the average latency, when the user number is less than 1600, the average latency of PDOA is the lowest and thus the deadline guarantee rate of PDOA is the highest. However, it reduces rapidly when the user number exceeds 1600. This is because the interference among users becomes severe and dominates the performance. The deadline guarantee rate of EGO is the highest with large user number. When there are 3000 users, the deadline guarantee rate is 93.2 percent while it is below 87 percent with the other three methods.

5.2 Impact of BS Number

The average energy consumption of the four methods with 3000 users is shown in Fig. 4, where the BS number varies from 35 to 77. It is found that with SOP, PDOA, DMDP and EGO, the average energy consumption decreases as the BS number increases. This is because with more BSs and more resources, the interference among users becomes weaker. Compared with SOP and EGO, the average energy consumption curves of PDOA and DMDP decay more rapidly. This is because PDOA and DMDP are more sensitive to the interference among users.

5.3 Impact of Time Slot Duration τ

The average energy consumption of the four methods with the time slot duration, τ, varies from 1 to 32 is shown in Fig. 5a. As can be seen from the figure, the value of τ has no impact on SOP since it takes no operation in each time slot. It is found that with PDOA, DMDP and EGO, as τ increases, service migration happens less frequently and thus the energy consumption of migration reduces. This leads to the reduction of average energy consumption. However, as τ exceeds a certain threshold (8 for PDOA, and 2 for DMDP and EGO), transmission energy consumption in request routing dominates the performance, thus the average energy consumption of PDOA increases.

The average service latency of the four methods is depicted in Fig. 5b. Again, we found that the value of τ has no impact on SOP due to the same reason described above. Similarly, with τ increases, the average service latency of PDOA, DMDP and EGO first decreases due to the reduction in service migration latency, and then increases after τ exceeds a certain number as transmission latency begins to dominate the performance. Taking the average energy consumption and service latency into account, we set $\tau = 2$ in our simulations.

5.4 Impact of User Mobility

We use the velocity of the trajectory as the standard (the average velocity of user is 1.4 m/s), and scale up/down the velocity of the users simultaneously to represent different levels of user mobility. The average service latency of the four methods with different user mobility, when the user number is 3000, is

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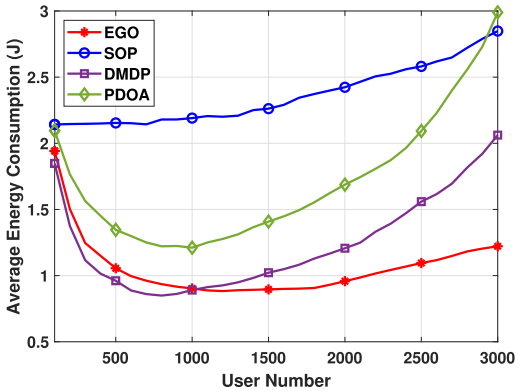
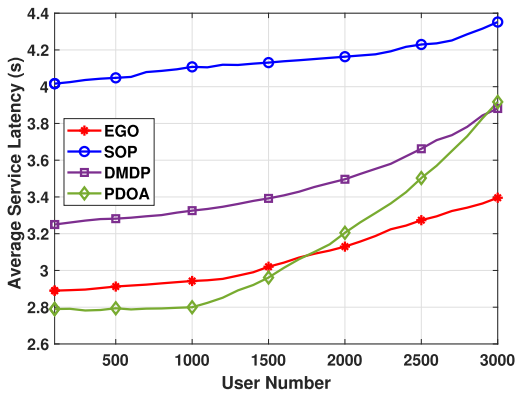
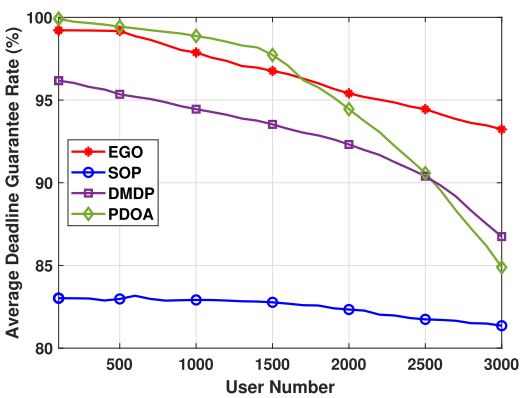


(a) The average energy consumption.



(b) The average service latency.



(c) The deadline guarantee rate.

Fig. 3. Average energy consumption, average service latency and average deadline guarantee rate of the four methods, where the user number varies from 100 to 3000.

depicted in Fig. 6a. We found that the average velocity of users has negligible impact on SOP since it takes no operation as users move. With the average velocity of users increases, the average service latency of PDOA, DMDP and EGO also increases. This is because the frequent service migration triggered by high user mobility aggravates the interference among users. It is also shown in Fig. 6b that the deadline guarantee rate of PDOA, DMDP and EGO reduces with the average velocity of users increases.

5.5 Impact of V and V'

Fig. 7a shows average energy consumption and average service latency of the EGO algorithm, where V varies from

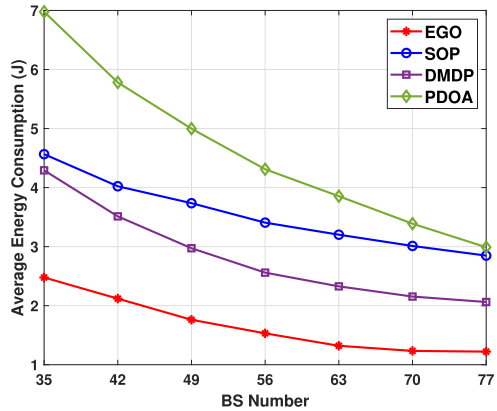
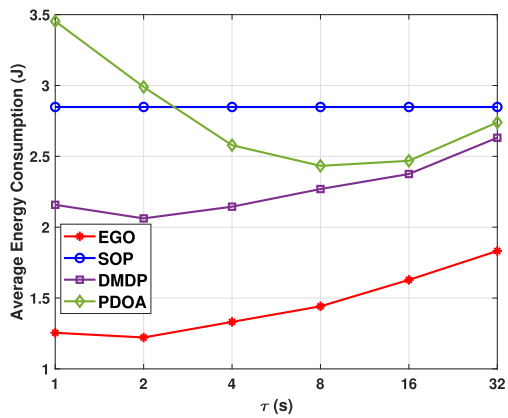


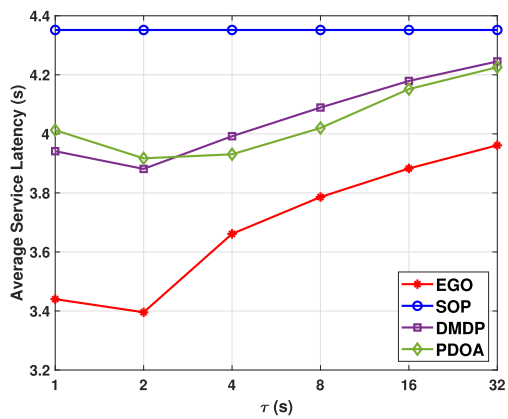
Fig. 4. Average energy consumption of the four methods with 3000 users, where the BS number varies from 35 to 77.

10^{-1} to 10^3. By increasing V from 10^{-1} to 10^3, EGO lays more emphasis on the average energy consumption than the resource utilization. It is found that as the average energy consumption decreases, the average service latency increases. By adjusting the value of V, we can achieve a balance between average energy consumption and average service latency. In our simulations, we set $V = 15$.

Fig. 7b shows the impact of V' on average energy consumption and average service latency. Similarly, as V' increases, EGO emphasizes more on average energy consumption than the energy consumption of migration. In this



(a) The average energy consumption.



(b) The average service latency.

Fig. 5. Average energy consumption, average service latency of the four methods with different 3000 users, where τ varies from 1 to 32.

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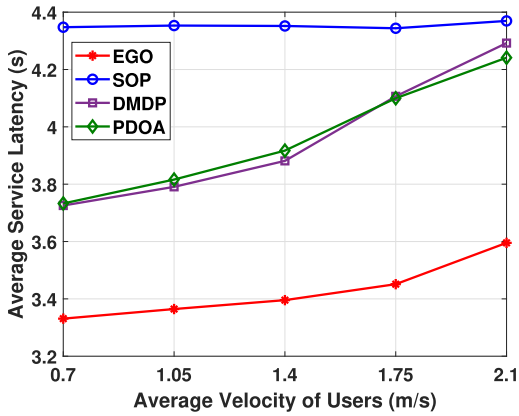
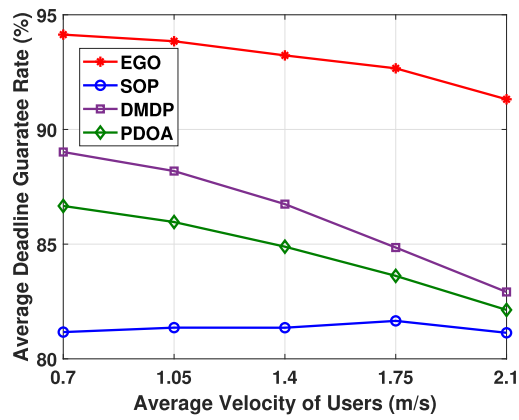
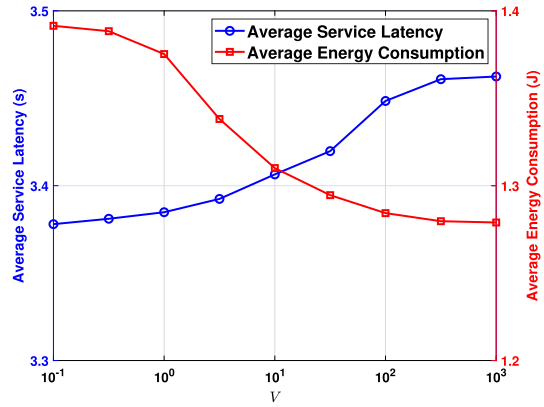


(a) The average service latency.

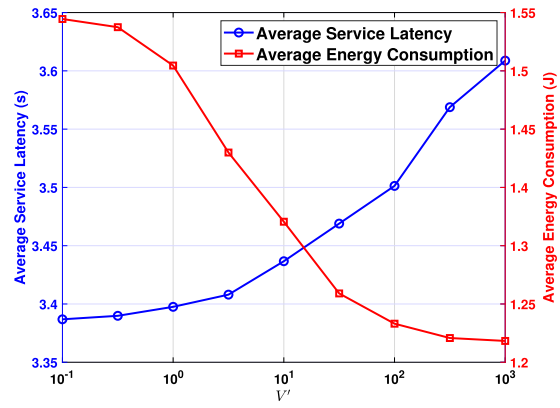


(b) The deadline guarantee rate.

Fig. 6. Average service latency and deadline guarantee rate of the four methods with different user mobility, when the user number is 3000.



(a) The impact of V.



(b) The impact of V'.

Fig. 7. Average energy consumption and average service latency of the EGO algorithm, where V and V' vary from 10^{-1} to 10^{3}.

case, service migration takes place more frequently, and thus the average service latency increases. By adjusting the value of V', we can achieve a balance between the average energy consumption and average service latency. In our simulations, we set $V' = 20$.

5.6 Convergence of the Modified PSO Algorithm

We show the convergence performance of the modified PSO algorithm with different J values in Fig. 8, where the particle number of each population is 20, i.e., $K = 20$. The values of ω, r_1, and r_2 are set as 1, 2 and 5, respectively. It is found that with smaller values of J, the modified PSO algorithm converges faster, e.g., it converges after 20 and 30 iterations with $J = 1$ and $J = 2$, respectively. However, the algorithm cannot converge to the optimal solution with small J values. On the other hand, with larger values of J, more iterations are required before the algorithm converged, e.g., 70 and 80 iterations are needed with $J = 4$ and $J = 5$, respectively. However, the improvement is negligible after J exceeds a certain threshold. In our simulations, we set $J = 3$.

5.7 Average Service Latency of Each Service

Besides the average service latency, we also show the average service latency of each of the 7 services in Fig. 9, where there are 3000 users. The red lines are the deadlines of each service. It is noticed that SOP performs the worst, i.e., the average service latency exceeds the response deadline for all the services. Again, this is because it does not migrate the service. It is also observed that the average service latency of DMDP and PDOA is above the response deadline of 5 services, due to the interference among users. In contrast, EGO satisfies the deadline requirements of all the services, even in low priority and data-intensive service, e.g., *platoon* and *face detection*.

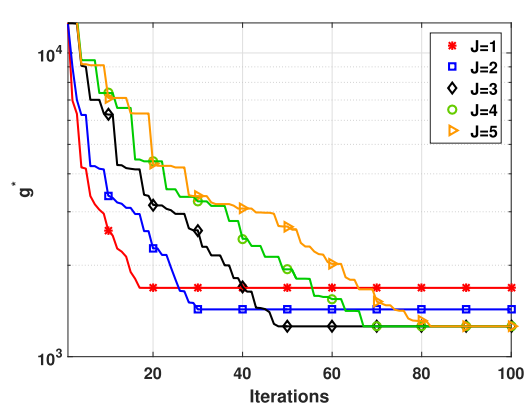


Fig. 8. Convergence of the modified PSO algorithm.

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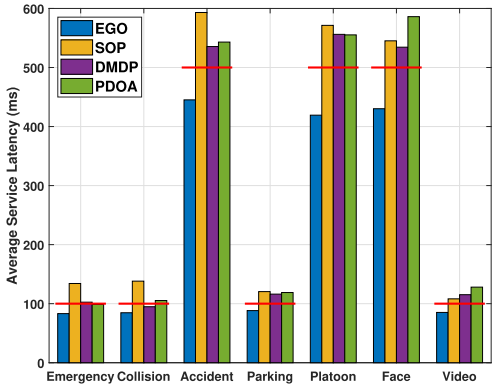

Fig. 9. Average service latency of each service with 3000 users.

6 Conclusion

In this study, we investigated the multi-user service migration problem in MEC-enabled heterogeneous dense cellular networks. We formulated the service migration process based on 3-layer virtual machine technique as a MINP optimization problem to minimize the average energy consumption with service latency requirements, where the interference among users was taken into account. An energy-efficient online algorithm, EGO, was proposed to solve the MINP problem. First, EGO used the Lyapunov optimization technique to transform the MINP problem into a series of per-service per-time slot sub-problems to avoid the interference among users and eliminate the need for user trajectory prediction. Meanwhile, a modified particle swarm optimization algorithm was proposed to converge to the optimal solution. We also proved that the solution of the EGO algorithm was similar to the optimal solution of the originally formulated MINP problem. The simulation results established with the real-world vehicle trajectory in Bologna demonstrated that the proposed solution achieved a performance gain of 40.7, 12.6 and 7.2 percent with large user numbers as compared to the state of the art solutions in terms of average energy consumption, average service latency and service deadline guarantee rate, respectively. This indicates that EGO was suitable for multiple users, especially in MEC-enabled heterogeneous dense cellular networks. It should be noted that in this paper, the mobile user can only connect to a single base station. In multi-access MEC where the mobile user can connect to multiple base stations at the same time, more sophisticated service migration algorithms are needed. This is left as a future study.

Acknowledgments

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